Levi Felix, PhD

Postdoctoral Research Associate
Department of Materials Science and Nanoengineering
Rice University
Houston-TX, United States

lf33@rice.edu
levifelix1@gmail.com
LinkedIn
Google Scholar
ORCID
Github

Education

2018 – 2022 **Ph.D., State University of Campinas - UNICAMP** in Physics.
Thesis title: *Computational Mechanics of Porous and Two-dimensional Carbon Nanomaterials.*

2016 – 2018 **M.Sc., Federal University of Ceará - UFC** in Physics.
Dissertation title: *Wave-packet propagation in phosphorene in the presence of grain boundaries.*

2013 – 2016 **B.S., Federal University of Ceará - UFC** in Physics (Magna Cum Laude).
Monography title: *Landau levels in graphene influenced by one dimensional potentials.*

Skills

Softwares LAMMPS, SIESTA, Quantum ESPRESSO, CP2K, VASP, DFTB+, VMD, VESTA and OVITO.

Coding Python, C++ and LaTeX.

Methods Molecular Dynamics Simulations, Density Functional Theory, Density Functional Tight-Binding, Nudged-Elastic Band, Umbrella Sampling, Metadynamics.

Awards and Honors

2022 **Recognition Award from NIST** for oral presentation on *Symposium DS02: Advanced Manufactured Materials—Innovative Experiments, Computational Modeling and Applications* at MRS Spring Meeting 2022 in Honolulu-HI, United States.

2021 **Best Poster Award** on *Symposium EQ20: Beyond Graphene 2D Materials Synthesis, Properties and Device Applications* at MRS Fall Meeting 2021 in Boston-MA, United States.

2016 **Magna Cum Laude** on *Bachelor of Physics* at Federal University of Ceará in Fortaleza-CE, Brazil.

Conferences and Proceedings

2022 **MRS Spring Meeting**. *Oral presentation*. Honolulu-HI, United States.

2021 **MRS Fall Meeting**. *Poster presentation*. Boston-MA, United States.

2019 **MRS Fall Meeting**. *Poster presentation*. Boston-MA, United States.

MRS Brazil Meeting. *Poster presentation*. Balneário Camboriú-SC, Brazil.

Employment History

Oct 2022 – Present **Postdoctoral Research Associate.** Rice University, Department of Materials Science and Nanoengineering

Aug 2019 – Dec 2019 **Graduate Teaching Assistant.** State University of Campinas (Unicamp), 'Gleb Wataghin' Institute of Physics. General Physics IV

Aug 2017 – Dec 2017 **Graduate Teaching Assistant.** Federal University of Ceará (UFC), Physics Department. Laboratory of Electricity and Magnetism I

Languages

Portuguese	Native
English	Fluent
Spanish	Intermediate

Publications

1. Felix, L. C., Pereira, M. L., Fabris, G. S., Fonseca, A. F., & Galvão, D. S. (2026). Kapitza resistance and thermal transport in knotted graphene nanoribbons. *Applied Surface Science*, 167187. doi:https://doi.org/10.1016/j.apsusc.2026.167187

2. Felix, L. C., Li, Q.-K., Penev, E. S., & Yakobson, B. I. (2025). Ab Initio Molecular Dynamics Insights into Stress Corrosion Cracking and Dissolution of Metal Oxides. *Materials*, *18*(3), 538. doi:10.3390/ma18030538

3. Felix, L. C., Ambekar, R., Tromer, R. M., Woellner, C. F., Rodrigues, V., Ajayan, P. M., … Galvao, D. S. (2024). Schwarzites and triply periodic minimal surfaces: From pure topology mathematics to macroscale applications. *Small*, 2400351. doi:10.1002/smll.202400351

4. Li, H., Felix, L. C., Li, Q., Ruan, Q., Yakobson, B. I., & Hersam, M. C. (2024). Atomic-resolution vibrational mapping of bilayer borophene. *Nano Letters*, *24*(34), 10674–10680. doi:10.1021/acs.nanolett.4c03224

5. Tromer, R. M., Felix, I. M., Felix, L. C., Machado, L. D., Woellner, C. F., & Galvao, D. S. (2024). Hydrogen atom/molecule adsorption on 2d metallic porphyrin: A first-principles study. *Chemical Physics*, *577*, 112142. doi:https://doi.org/10.1016/j.chemphys.2023.112142

6. Tromer, R. M., Felix, L. C., Baughmann, R. H., Galvao, D. S., & Woellner, C. F. (2024). Transforming two-dimensional carbon allotropes into three-dimensional ones through topological mapping: The case of biphenylene carbon (graphenylene). *The Journal of Physical Chemistry A*, *128*(35), 7346–7352. doi:10.1021/acs.jpca.4c01339

7. de Alcântara, A. C. S., Felix, L. C., Galvão, D. S., Sollero, P., & Skaf, M. S. (2023). The role of the extrafibrillar volume on the mechanical properties of molecular models of mineralized bone microfibrils. *ACS biomaterials science & engineering*, *9*(1), 230–245. doi:10.1021/acsbiomaterials.2c00728

8. Moreira, M., Felix, L. C., Cottancin, E., Pellarin, M., Ugarte, D., Hillenkamp, M., … Rodrigues, V. (2023). Influence of cluster sources on the growth mechanisms and chemical composition of bimetallic nanoparticles. *The Journal of Physical Chemistry C*. doi:10.1021/acs.jpcc.2c08044

9. Felix, L., Ambekar, R. S., Woellner, C., Kushwaha, B., Pal, V., Tiwary, C. S., & Galvao, D. S. (2022). Mechanical properties of 3D-printed pentadiamond. *Journal of physics D: Applied physics*. doi:10.1088/1361-6463/ac91dc

10. Felix, L. C., & Galvao, D. S. (2022a). Guided fractures in graphene mechanical diode-like structures. *Phys. Chem. Chem. Phys.* doi:10.1039/D2CP01207C

11. Alcântara, A. C. S., Felix, L. C., Galvão, D. S., Sollero, P., & Skaf, M. S. (2022). Devising bone molecular models at the nanoscale: From usual mineralized collagen fibrils to the first bone fibers including hydroxyapatite in the extra-fibrillar volume. *Materials*, *15*(6). doi:10.3390/ma15062274

12. Felix, L. C., Tromer, R. M., Woellner, C. F., Tiwary, C. S., & Galvao, D. S. (2022a). Mechanical response of pentadiamond: A DFT and molecular dynamics study. *Physica B: Condensed Matter*, *629*, 413576. doi:10.1016/j.physb.2021.413576

13 Gaal, V., Felix, L. C., Woellner, C. F., Galvao, D. S., Tiwary, C. S., d'Ávila, M. A., & Rodrigues, V. (2021). Mechanical properties of 3D printed macroscopic models of schwarzites. *Nano Select*. doi:10.1002/nano.202100147

14 Tromer, R. M., Felix, L. C., Ribeiro, L. A., & Galvao, D. S. (2021). Optoelectronic properties of amorphous carbon-based nanotube and nanoscroll. *Physica E: Low-dimensional Systems and Nanostructures*, *130*, 114683. doi:10.1016/j.physe.2021.114683

15 Tromer, R. M., Felix, L. C., Woellner, C. F., & Galvao, D. S. (2020a). A DFT investigation of the electronic, optical, and thermoelectric properties of pentadiamond. *Chemical physics letters*, 138210. doi:10.1016/j.cplett.2020.138210

16 Cunha, S. M., da Costa, D. R., Felix, L. C., Chaves, A., & Pereira, J. M. (2020). Electronic and transport properties of anisotropic semiconductor quantum wires. *Phys. Rev. B*, *102*, 045427. doi:10.1103/PhysRevB.102.045427

17 Felix, L. C., Gaál, V., Woellner, C. F., Rodrigues, V., & Galvao, D. S. (2020). Mechanical Properties of Diamond Schwarzites: From Atomistic Models to 3D-Printed Structures. *MRS Advances*, *5*(33), 1775–1781. doi:10.1557/adv.2020.175

18 Felix, L. C., Tromer, R. M., Autreto, P. A. S., Ribeiro Junior, L. A., & Galvao, D. S. (2020). On the mechanical properties and thermal stability of a recently synthesized monolayer amorphous carbon. *The Journal of Physical Chemistry C*, *124*(27), 14855–14860. doi:10.1021/acs.jpcc.0c02999

19 Tromer, R. M., Felix, L. C., Woellner, C. F., & Galvao, D. S. (2020b). On the structural stability and optical properties of germanium-based schwarzites: A density functional theory investigation. *Physical Chemistry Chemical Physics*, *22*(28), 16286–16293. doi:10.1039/d0cp02143a

20 Felix, L. C., Woellner, C. F., & Galvao, D. S. (2019a). Mechanical and energy-absorption properties of schwarzites. *Carbon*. doi:10.1016/j.carbon.2019.10.066